CS315 · ALGORITHMS ANALYSIS & DESIGN

تدريب Midterm 472

ميد 472 · من L1 إلى L7

محارب

01 عن الميد

هذي نسخة معاد كتابتها لميد 472، مكتوب عليها «From L1 to L7». الأسئلة منقولة مثل ما هي. ما فيها توزيع درجات.

مهم

ما عندي نموذج إجابة رسمي لهذا الميد. الحلول من حلي، وكل الجداول انحسبت بكود. أسئلة الـ Matrix chain و LCS و B-tree من محاضرات ما وصلتني سلايداتها، فحليتها بالطريقة المعروفة في الكتاب. لو طريقة الدكتور في السلايدات تختلف، امش على طريقته.

الأسئلة 02

MATRIX CHAIN Q1

Apply the Dynamic programming algorithm and give the optimal solution to the matrix chain multiplication ABCD where the order for matrix:

A = 4 × 5 B = 5 × 8 C = 8 × 3 D = 3 × 7

Minimum operations = _____ Best order of multiplication: _____

	A	B	C	D
A	0	160		
B		0	120	
C			0	
D				0

TRUE / FALSE + MCQ **Q2**

1. The code length in Huffman codes depend on the frequency of occurrence of characters. ○ True ○ False

2. Greedy algorithm can solve all the dynamic programming problems. ○ True ○ False

3. The time to run MERGESORT on a list of size n, $T(n) = T(\frac{n}{2}) + T(\frac{n}{2}) + T(MERGE(\frac{n}{2}))$ ○ True ○ False

4. In order to compare two algorithms, the same hardware and software environments must be used without any consideration of inputs type and its size. ○ True ○ False

5. Theoretical analysis allows us to evaluate the speed of an algorithm independent of the hardware/software environment. ○ True ○ False

6. An algorithm which does NOT use the past results and uses them to find new results is:

a. Greedy Method b. Divide and conquer c. Dynamic programming d. None of the mentioned

LCS **Q3**

Using the dynamic programming approach, find the longest common subsequence between two strings:

X = ACBAED and Y = ABCADF.

Length _____ The subsequence: 1 _____ 2 _____

	⌀	A	B	C	A	D	F
⌀							
A							
C							
B							
A							
E							
D							

HASH TABLE **Q4**

Insert integer keys A = {4, 13, 49, 6, 40, 33} in order into a hash table of size 9 using the hash function h1(k) = k mod 9. Collisions should be resolved via chaining, where collisions are stored at the end of a chain. Fill in the hash table after all keys have been inserted.

0	1	2	3	4	5	6	7	8

B-TREE **Q5**

A) Create a B-tree with order 5, min degree 2 the following item [24, 52, 90, 40, 11, 86, 61, 50]

B) insert 29

C) delete 52, 11